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A comprehensive review of virtual synchronous generator

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ABSTRACT

The renewable energy resources based distributed generation (DG) unit's integration into the power system is growing rapidly, and the power system experiencing low inertia and damping, which causes severe stability and performance issues of the power system. In order to facilitate the integration of DG into the existing power system, improved control schemes for grid-tied electronic converters are necessary to ensure the stability and support to the power system. The virtual-synchronous generator (VSG) control emulates the dynamics of the rotation synchronous generator and enhances the stability of the power system. In this paper, an overview of different topologies to virtual inertia is presented along with a detailed description of the VSG structure. Moreover, VSG control defining active and reactive power control, and voltage and frequency control is explained in detail. The VSG stability analysis methods, such as small signal stability and transient stability are presented in comprehensive manners. Later on, the VSG application fields are described. Finally, the VSG problems and future research directions such as centralised control, stability improvement, reliable modelling, the interface for VSG are discussed.

1. Introduction

The stocks of conventional fossil energy have been depleted significantly and unable to meet the needs of the sustainable development of human society. The advantages of renewable energy resources based distributed generation are flexibility, pollution-free, and wide distribution. It becomes a suitable energy that conforms to the concept of sustainable development [1,2]. In recent years, a large number of renewable energy-based distributed power generation units such as photovoltaic and wind farms are integrated into conventional power system [3,4].

The DGs consisting photovoltaic systems, there is no rotating part of the inertial response, and it can participate in frequency support by adding virtual inertia via electronic inverters, whereas in conventional power generating units, synchronous generator (SG) provides the frequency support during the disturbances via its rotating mass. However, the DGs which are electronically interfaced with power system or grid, demonstrate different characteristics than conventional power generating units. In electronically interfaced DGs, the power generated at the primary side of DGs is regulated by electronic inverters, but they are not able to supply the required inertia and damping to the power grid [5–9].

Consequently, electronically interfaced DGs cannot improve the system stability. However, the solution of this problem is developed by applying the suitable control techniques to the grid connected inverter and controlling its switching pattern so that it works as an SG by mimicking the behaviour of SG. The grid connected inverters, which mimic the steady-state and transient characteristics of SG, are called VSGs (virtual synchronous generators). It is predicted that VSG integrated systems will be the future of power system network [10–12]. Therefore, it is utmost essential to analyse and improve the transient stability of VSG based power systems [13].

The basic idea of VSG with different approaches is presented in [14–21], which can make the grid connected the inverter to mimic the operating characteristics of SG. Moreover, the operational characteristics of conventional SG and VSG can also be compared, such as Q-δ, P-f [22]. Consequently, in [14], the proposed VSG scheme mimics the swing equation of conventional SG, while the proposed technique in [17] reflect the proper operating characteristics of the synchronous machine. However, current-controlled VSG is equivalent to the current source, so it is challenging to provide voltage and frequency support for the system. Therefore, to overcome the deficiencies of current-controlled VSG, a voltage-controlled VSG technique is proposed in [23]. The essence of voltage-controlled VSG techniques is to simulate the rotor inertia and system frequency modulation characteristics of SG in frequency control, to improve the frequency stability of the system; at the same time, in voltage control, the reactive voltage relationship is mainly considered to control the stable voltage output [24]. The power controller and voltage frequency controller make VSG perform functions of power control and frequency modulation [25,26].

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Additionally, several techniques are demonstrated in [19,27–30], which develops the model and control to mimic the various control dynamics of SG. These model and control techniques achieve the working equivalence and self-synchronisation of SG without phase-locked loop (PLL). Therefore, VSG control is still needed to be mature enough in theoretical and practical application in comparison to a conventional SG [31].

The currently available literature possesses the basic and improved ideas of different VSG control techniques, modifications in VSG control to improve functionality, and its implementation. However, being a review article, this paper first introduces the different topologies to produce virtual inertia in Section 2. The VSG basic principle and control orders are discussed in Section 3. In Section 4, the VSG operation control is discussed in detail. The VSG stability analysis is explained in Section 5. Section 6 describes the VSG applications. Section 7 briefs about the VSG problems and future research directions and Section 8 presented the conclusion of this paper.

2. Introduction of current virtual inertia topologies

VSG is a combination of control algorithms, renewable energy sources, energy storage systems, and power electronics that emulates the inertia of a conventional power system [32]. VSG algorithm is the primary part of the system which interfaced among different storage units, generation units and the utility grid. Variable wind turbines are used in modern-day grid systems, and these turbines are connected with back to back inverters which provide complete decoupling of inertia from the utility grid. Energy storage system and photovoltaic systems interfaced via DC to DC converters and an additional inverter at the front end. This system does not respond to inertia changes [33]. According to literature, the primary model concepts are similar for different topologies; however, implementation of each topology model is different from others. Few topologies use mathematical equations to fully mimic synchronous generator behaviours whereas few topologies use swing equation to copy irregular performance of the synchronous generator. In a few topologies, DG units react to the frequency changes of the utility grid system. Various important topologies of VSG are discussed in this section.

2.1. Synchronverters

A synchronverter is equivalent to an SG with a small capacitor bank connected in parallel to the stator. The dynamic equations are the same; only the mechanical power exchanged with the prime mover is replaced with the power exchanged with the dc bus [34]. This topology does not require a significant structural change in the design and power system to perform its operations traditionally. Q.C.Zhong developed this strategy very well in [35]. Similar to the synchronous generator output power regulation, the output power of the inverter is regulated by frequency droop algorithm [36].

The structure of a synchronverter is shown in Fig. 1. The differential equation of the controlling unit is solved by the output voltage and current signals of the inverter. Moreover, the damping factor and moment of inertia can be adjusted according to needs. However, considering the stability of the system, these parameters are very crucial [37].

2.2. Topology developed in ISE lab

In this topology, the swing equation for power-frequency is used in place of a complete synchronous grid model [38]. A simple model of this topology is shown in Fig. 2.

Frequency and power measurement unit take the output current of inverter and gather the voltage from the common connection point. By these two values, it calculates the inverter active output power as well as utility grid frequency. Prime mover input power is applied to the

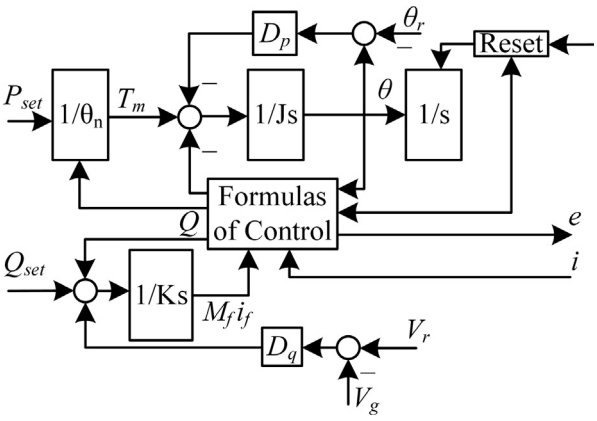


Fig. 1. Simple structure of Synchronverter topology [19].

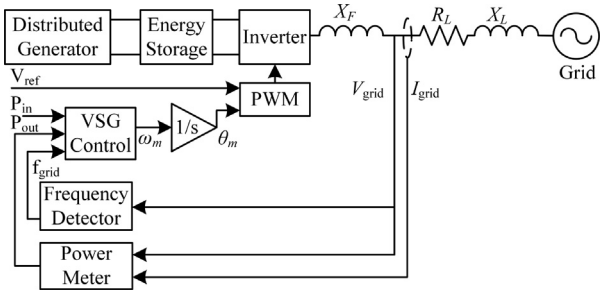


Fig. 2. Simple structure of ISE topology [39].

control algorithm unit along with two calculated values [40].

2.3. KHI topology

This topology is developed by Kawasaki Heavy Industries (KHI) lab. An automatic voltage regulator and a governor unit constructed on a digital controlling unit to produce the reference phase and voltage of the virtual machine [16]. Later on, these references are used to produce reference currents for a synchronous generator according to algebraic phasor representation. A simple model of KHI lab topology is shown in Fig. 3.

2.4. VISMA and IEPE topologies

Few other topologies and techniques are introduced in different research work such as Virtual synchronous machine (VISMA) and Institute of electrical power engineering (IEPE) technique [41,42]. The fundamental idea to mimic inertia response is uniform for all topologies. The VISMA technique used d-q based architecture to mimic the synchronous generator in the research work of [43]. When this architectural design applied through a digital control unit of power inverter, then it copies the dynamics of synchronous generator. However, in literature, it is discussed that VISMA technique is unstable due to numerical data involvement. In [44], instead of d-q model, a new

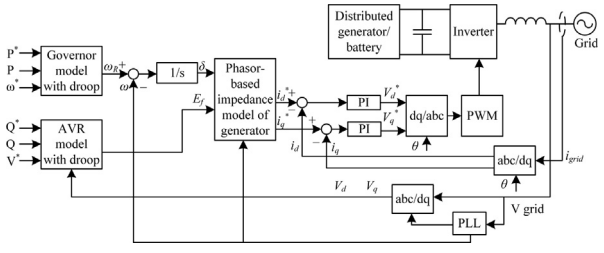


Fig. 3. Simple structure of KHI topology [39].

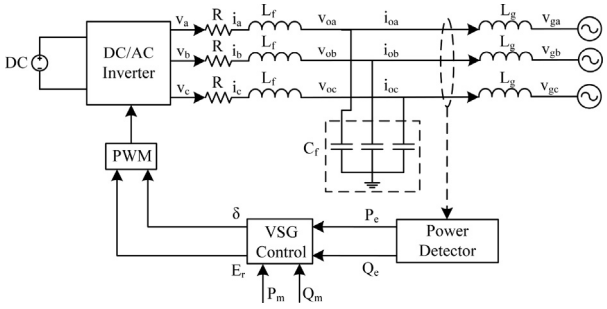


Fig. 4. Simple VSG Structure.

approach is developed to increase the strength by using a three-phase model. This new technique is highly efficient for asymmetrical loads and sharp changes in the utility grid.

IEPE topology is another topology which is somewhat similar to VISMA, but the significant difference is that IEPE used output current of a DG and then produce reference voltage for virtual machines. In grid connected mode of IEPE, it is problematic to cope with transient currents in the synchronization period. However, for islanded mode, IEPE technique is best fitted.

3. VSG principles and control orders

A basic and pretty simple structure of VSG is shown in Fig. 4, and it can be observed that VSG consist of a DG unit, energy storage device, DC/AC converter, a filter circuit, governor and grid. If the power of the distributed generator and energy storage system is assumed as the input torque of the prime mover, while DC/AC converter is assumed as the electromechanical energy transformer between stator and rotor. Then, the fundamental component of midpoint voltage is representing the electromotive force of the VSG. Resistance and inductance of filter unit representing the stator winding impedance.

According to Fig. 4, v_{abc} and i_{abc} are the AC side voltage, and current of the inverter, respectively. v_{oabc} and i_{oabc} are the LC filtered voltage, and current, respectively. v_{gabc} is the grid voltage, and R, L_f, and C_f are the filter resistance, inductance, and capacitance, respectively. L_g is the gridline inductance, E and δ are the internal potential amplitude and phase angle of VSG, V and θ is the voltage amplitude and phase of VSG terminal, and P_e, P_m and Q_e, Q_m are the active power and reactive power output by VSG.

The higher-order or lower-order VSG control strategies are those which can be developed by minor modification in the control structure of the voltage source controller, as illustrated in Fig. 5. Although both types of VSG strategies provide active and reactive power control, however, each strategy has a specific control frame. Moreover, various VSG control strategies have been developed to enable an electronic convertor to mimic the characteristics of an SG [45,46].

The simple swing equation of SG is used as a core part of VSG, and it

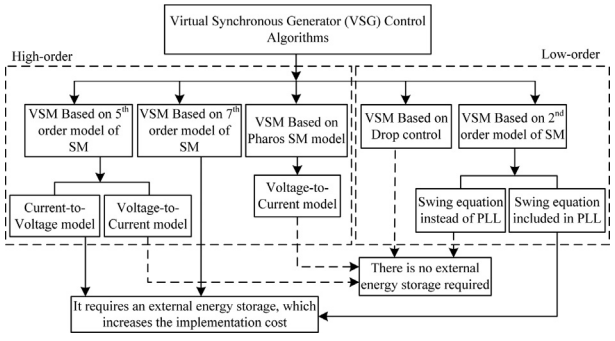


Fig. 5. Classification of VSG based on model's order [46].

is stated as [19,40,46]:

$$J\frac{d\omega}{dt} = T_m - T_e - D(\omega - \omega_r)\frac{d\delta}{dt} = \omega \tag{1}$$

where ω and ω_r are virtual angular frequency and reference angular frequency, respectively. T_m and T_e are mechanical torque and electromagnetic torque, respectively. δ, D, and J is the power angle, damping coefficient, and rotor moment of inertia, respectively.

Moreover, to mimic the electromagnetic characteristics of SG for VSG, the stator electrical equation of the synchronous generator is generally simulated without considering the electromagnetic relationship between stator and rotor and it can be expressed as [19,46]:

$$L_f\frac{di_{abc}}{dt} = e_{abc} - v_{abc} - Ri_{abc} \tag{2}$$

In order to calculate the reference phase and frequency of modulating signal, the active power loop of VSG mimics the primary frequency regulation, damping and inertia of SG. Whereas, modulating signal amplitude is calculated by reactive power loop, which mimics the voltage regulation of SG. The general and simple swing equation of SG is used as a core part of VSG, and it is stated as [19,46]:

$$P_m - P_e = \left(2J\frac{d\omega}{dt} - D(\omega - \omega_r)\right)\omega \tag{3}$$

where P_m is the inverter input power, P_e is the output electrical power of the inverter. In order to keep the VSG speed equal to the grid frequency, virtual damping coefficient plays an important role.

The mathematical equation for active power loop of VSG, including the simple governor of VSG is described as [19,46]:

$$J\frac{d\omega}{dt} = \frac{P_m}{\omega} - \frac{P_e}{\omega} - D(\omega - \omega_r) \tag{4}$$

Moreover, the mathematical equation for reactive power loop of VSG can be stated as [19,46]:

$$K\frac{dE_r}{dt} = Q_m - Q_e + k_q(V_r - V) \tag{5}$$

where K is the reactive power-voltage inertia coefficient of the reactive power loop, E_r is the virtual electromotive force, Q_m is the reference reactive power, Q_e is the reactive power output; k_q is the reactive power-voltage droop coefficient; V is the output voltage amplitude, and V_r is the rated voltage amplitude.

Consequently, the VSG models can be divided based on the requirement of an extra component such as energy storage; therefore, Fig. 5 summarises the differences of VSG models at different stages, and they were found in the various literature [19,46,47].

4. VSG operation control

The input/output control of VSG is achieved by the suitable working mechanism of the inverter at its interface. The common control methods are active and reactive power control, voltage and frequency control, and droop control. In general, the application of the control strategy into the system depends upon the operational mode of the power system.

A typical invertor and a VSG are somewhat similar; however, the only difference is that the VSG mimics the characteristics of an SG through the control algorithm. The control algorithm of VSG can be divided into two broad categories, which are described as following.

4.1. Active and reactive power controls

Active and reactive power control makes distributed generators' output a constant scheduling power, and it consists of the outer power loop and inner current loop. This control technique is widely used in grid connected VSG units. The converter underactive and reactive

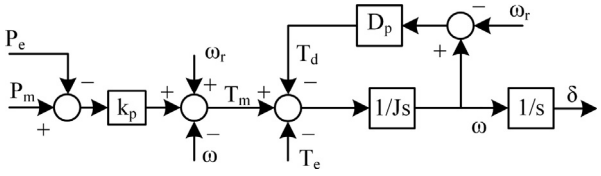


Fig. 6. VSG control of active power and frequency [48].

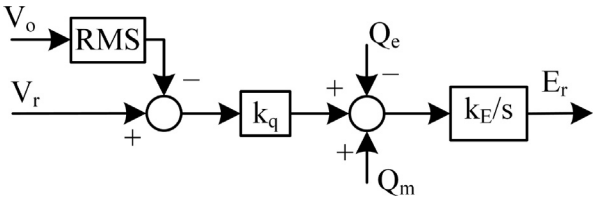


Fig. 7. VSG control of reactive power and voltage [48,63].

power control acts as a control bus in the power system. These control buses account for most of the buses in the power grid system. Generally, the common storage devices and electric vehicles are active and reactive control buses.

When the active and reactive power technique is applied in grid connected mode, active power could follow the dispatch command very well by the governor unit. However, the reactive power controller does not follow the dispatched command effectively and fail to supply the required reactive power to the system. The active power control of VSG is a replica of the governor unit of SG. The control diagram is shown in Fig. 6, where, P_e and P_m is the VSG active power and reference active power, respectively.

Currently, most VSG control techniques use the active and reactive power control method because of its simplicity [49–54]. It can realise the active power distribution according to the capacity, especially when multiple VSGs are operating in parallel. During grid connected mode, reactive power controller unit fails to deliver the required power; therefore, it is necessary to incorporate a unit to improve the functionality of the reactive power unit. In [55,56] an improved reactive power controller is proposed which supply reactive power to the power system by following the dispatched command.

Considering the accuracy of active power distribution, the [57] added an auxiliary active and reactive power term to the active power loop. This additional term ensures steady state characteristics and improves the dynamic characteristics. Moreover, it helps to suppress the negative influence of the line impedance between VSGs on the active power characteristics. Therefore, this modification improves the VSGs operation in a parallel configuration.

Moreover, in comparison to active power control, the reactive power control is easily affected by the line impedance, load fluctuation, and other factors, which makes the control result deviate from the required characteristics, and ultimately leads to the inaccuracy of reactive power distribution. To suppress the impedance and load fluctuation effect, various control modifications are proposed in [58–60]. For example, the adaptive parameter estimation and selection technique, and the virtual impedance control method is an introduction to reduce the output voltage. These techniques are useful to control and gain better effects for reactive voltage control of VSG.

Moreover, in [48], the active and reactive power control is discussed by parameter estimation for improving the economical operation of VSG. The damping effect and inertia is considered as a constant. However, for parameter adjustment the minimum energy storage capacity is used as a reference, which reduces the cost of active power control of VSG.

The reactive power is difficult to be accurately distributed due to the close coupling of active and reactive components during grid connected operation. Therefore, in [33], the traditional power decoupling mechanism is initially analysed, which indicates that the traditional power decoupling method is unsuitable for medium- and low-voltage. Thus, an improved power decoupling method is proposed, which estimate the voltage at the point of common coupling and tracking their reference values, the output active and reactive power of inverters can perform dynamic decoupling. Furthermore, the stability of the new control structure and selection of relevant coefficients are analysed in this research.

4.2. Voltage and frequency control

Voltage and frequency control can output constant voltage and frequency composed of the voltage outer loop and the current inner loop. This control technique is widely used islanded VSG units. During the islanding operation, the primary control unit using active and reactive power control provides voltage and frequency support for the passive unit as a slack bus. This technique possesses the high-speed characteristic which idealises the application of this technique in power generation or energy storage systems of photovoltaic and wind. The voltage control unit of the VSG achieves the voltage amplitude by mimicking the function excitation regulator of the SG. Generally, the voltage control adjusts the output voltage according to the VSG output voltage amplitude deviation and uses the voltage adjustment coefficient k_q to characterise the voltage regulation capability of the VSG [61,62]. The primary control structure of voltage control along reactive power control is shown in Fig. 7.

Considering Fig. 7, V_o and V_r are the actual voltage and the reference voltage, respectively. Q_e and Q_m are the calculated reactive power and reference reactive power, respectively. k_q and k_E are the droop coefficient and an integral coefficient of reactive voltage, respectively. E_r is the reference output voltage amplitude. Generally, the reference output voltage amplitude is combined with the angle of active frequency control to produce a VSG reference voltage.

In order to achieve the constant output voltage of VSG, [64] proposes a method for multiple VSG connecting in parallel. The parallel connected VSG generate an error signal when active and reactive power mismatch than the required power. This error signal is used to achieve the constant output voltage of VSG. Moreover, under-damping is another critical issue for VSG based power systems. To suppress this deficiency of VSG based power systems, [65] proposed a non-linear damping controller. This damping controller consists of an additional excitation controller which stabilise the VSG output voltage according to the reference voltages. .

Moreover, in the grid connect mode, the VSG frequency control issue is considered in [66,67] when the power system experience fault. The proposed VSG control provides effective frequency support when system experience the fault or disturbance. Consequently, the proposed control rapidly recovers the system frequency after fault is clear. Furthermore, [68] proposed an adaptive load shedding scheme to stabilise the VSG based power system. This proposed scheme locally measures the system parameters especially voltage and frequency signals. After that, a stepwise load-shedding will take place in locations where the highest voltage drop and frequency variation are experienced. Consequently, the load-shedding step changes according to certain factors such as shedding speed, location and value, and the rate of frequency change. The proposed approach eliminates the adjustable loads to return the frequency back to the desired value.

The unbalanced and distorted grid connection is another type of critical issue in conventional and VSG based power systems. Usually, when the grid voltage is unbalanced, the negative sequence voltage components will appear. The traditional VSG control strategy does not consider the negative sequence components and cannot eliminate them; consequently, it will lead to current unbalance and power oscillations. To address these problems, the various researchers proposed the different techniques to suppress these effects [69–71]. For example, in

[69], the comprehensive control strategy of VSG under unbalanced voltage conditions is proposed, which is capable to balance injected currents to reduce delivery of active power fluctuations and to suppress reactive power oscillations. This control technique also improved the low-voltage ride-through capability of VSG during grid faults. However, the results presented in this research are for single VSG connected with grid, whereas other different scenarios such as multiple VSG connect with grid or parallel integrated VSG and SG connect with the grid are not considered in it.

Self-Adaptive VSG is a relatively new control technique to provide frequency support in the VSG power system. The conventional VSG based power system has constant inertia, and the selection of different inertia values significantly affects frequency stability. To suppress these issues, the self-adaptive VSG technique is proposed in [72,73]. Therefore, in [73], a self-adaptive virtual inertia control system using fuzzy logic for ensuring stable frequency stabilization is proposed. In this control technique, the virtual inertia constant is automatically adjusted based on input signals of real power injection of renewable energy sources and system frequency deviations, avoiding unsuitable selection and delivering rapid inertia response. The simulation results show a significant improvement in frequency stability by implementing self-adaptive VSG; however, the paper lacks the experimental result. Consequently, simulation is results presented only for an islanded micro-grid, while results for multiple VSG based islanded microgrid are not discussed.

5. VSG stability analysis

The main goal of VSG to improve the stability of the power system with significant integration of renewable energy resource-based power electronics equipment. The power system stability or synchronous stability of a power system can be of several types depending upon the nature of the disturbance and for successful analysis. Generally, two types of stabilities as small signal stability and transient stability are considered for a VSG based power system. This section covers the small signal stability and transient stability of VSG.

5.1. Small signal stability analysis

Small signal stability refers to the power system able to maintain steady state voltage when subject to small perturbations such as incremental changes in system load. Generally, small signal stability is influenced by the characteristics of the load, continuous control and discreet control at a specific time [74,75]. Moreover, a lot of researcher work on small signal stability of VSG based power systems in grid connected and islanded mode.

In [76], the small-signal sequence impedance models of the voltage-controlled VSG and the current-controlled VSG are built, and the sequence impedance characteristics of these two types of VSGs are compared and analysed. The sequence impedance of the voltage-controlled VSG, being consistent with the grid impedance characteristics, is generally inductive. Based on the built sequence impedance models and the impedance stability criterion, the influence of the weakness of the grid on the stability of these two types of VSG grid-connected system is analysed. Moreover, the results of the stability analysis show that the current-controlled VSG is prone to instability while the voltage-controlled VSG remains stable. Therefore, the voltage-controlled VSG is more suitable than the current-controlled VSG for grid-connected renewable energy generation in an ultraweak grid from the point of view of system stability. However, researchers did not propose a solution to improve the stability of current-controlled VSG.

For the stability analysis of the parallel VSGs system with the proposed self-adaptable reactive power voltage controller, a linearised small-signal model is established, and the parametric sweep results of line impedance are analysed in [77,78]. It is considered that active power-frequency and reactive power-voltage controllers play a significant role in the VSG control system. However, the line impedance and local loads are the critical cases impacting on the control performance of the two controllers. Therefore, a self-adaptable reactive power-voltage controller is proposed to deal with the sharing problem of reactive power in the parallel VSGs system, which uses the reactive power difference to adjust reactive power voltage control coefficient and improves the stability of the system. Moreover, the stability analysis test is performed on only two systems while the authors did not give the stability analysis results for multiple parallels connected VSGs.

An enhanced controller of VSG without a PLL and the detailed small-signal modelling of the VSG control system is proposed in [79]. A linearised small-signal model of VSG in isolated operation is established in detail including electrical parts and control parts, which analyse dynamic characteristics of the enhanced VSG control system. The accuracy of the proposed small-signal model was assessed through a comparison with the results from the electrical simulation. The impact rules of the main parameters on VSG system have been proposed via small signal stability analysis, which helps to identify possible feedback signals for controller design to improve the system stability. However, the authors did not provide practical results for the proposed system. Moreover, the results for grid connected VSG systems are skipped in this research too.

Moreover, for VSG based wind turbines in [80] proposed a robust grid synchronisation method. This method employs and modifies the idea using the capacitor voltage of the converter to fulfil the PLL-less grid-synchronization. Moreover, virtual capacitor control and the stabilisation control are proposed simultaneously in order to improve the capability of inertial response and the stable operation in weak grids. Consequently, validation is conducted, and the overall system stability is evaluated through small signal analysis by using eigenvalue along with mechanism analysis of instability. However, authors postponed the research of internal frequency lock and non-communication based inertial system for the future.

Furthermore, an adaptive virtual inertia control strategy based on an improved bang-bang control strategy for a micro-grid is presented in [63,81,82]. This proposed strategy can use the variability of virtual inertia to reduce dynamic frequency deviation. Whereas, the steady-state interval of frequency and the steady-state inertia are set to improve the system frequency stability. Moreover, the stability analysis of the value range of the virtual inertia is performed by the small signal model of the VSG for the micro-grid. Simulation results support the proposed modification and analysis whereas experiments results are not included in this research article. Only the relationship between active power and frequency is discussed in this paper, while the relationship between reactive power and voltage is postponed for the future.

5.2. Transient stability analysis

The power system is continually being subjected to changes or disturbances of various forms: faults, load changes, connection/disconnection of power generators units. The power network is a complex electro-mechanical system, and these events cause oscillations in the speed and angles of machines and power flows along the lines. Transient stability analysis is the study of the system in response to these changes and is used to determine if the system will be stable after experiencing a disturbance. In comparison to small-signal stability analysis, transient stability analysis is generally challenging to linearise [53,83]. Consequently, for proper operation of the system, it is essential to ensure that after a disturbance, the system settles down to a new, stable condition [84].

At present, for transient stability analysis of VSG based power system, Lyapunov function technique, the centre of inertia technique and equal area criterion are the main techniques [85–89]. In [85], transient angle stability of a VSG is investigated by Lyapunov's direct method. The deteriorative effect of reactive power control loop on transient angle stability is analysed and voltage variation is

incorporated into an approximate Lyapunov's direct method. According to the proposed method, the inverter internal voltage is treated as a parameter rather than a state variable. Moreover, the influence of different parameters on transient angle stability is carried out along with an enhanced control strategy to improve the transient angle stability by adjusting the reference power. Similarly, in [86,90,91] various authors use the Lyapunov method to analyse the transient stability of VSG along with some modified and improved controls to improve the functionality of the power system.

In [87,88], the centre of inertia technique is used to analyse the transient stability of VSG while connecting with SGs, and modifications are proposed in the governor block to enhance transient stability. In [88] author proposed the modification to improve the transient stability and analyse it according to the centre of inertia reference frame. In this research, the author places a VSG with multiple SGs in 30-bus two area system. Considering the simulation results of the 30-bus two area system, it is observed and concluded that when a VSG integrated into 1 area, the SGs in the same area receive a positive effect of VSG integration. However, the effect of VSG in the second area is quite low.

In [92] the process of frequency oscillation during disturbance is theoretically analysed using power-angle curve and equal area criterion method, with the conclusion that the deviation and changing rate of angular velocity can be changed by inertia time constant of VSG. Similarly, [93,94] analyses the power-angle characteristic of VSG in stable operation and output current saturation condition, and then utilises the equal-area criterion used in synchronous generator stability analysis to calculate the fault critical clearing angle. However, these researches are limited to simple VSG based power system in grid connected mode, while skipping the analysis for islanded mode and Multiple VSG connected in parallel. Moreover, experimental results are not included in these researches. In [95], the VSG control algorithm adaptive virtual rotor inertia is proposed, and it is concluded that the inertia coefficient should be selected flexibly considering the requirements of transient overshoot response.

6. VSG applications

6.1. Photovoltaic & wind power systems and VSG

The photovoltaic and wind power system is an interconnection of the domestic and large scale of low voltage distributed energy sources. Generally, photovoltaic and wind power systems are placed in the category of low voltage and medium voltage distribution power systems. When few photovoltaic or wind power systems are connected with each other or with grid, it experiences specific difficulties such as power quality and power system stability.

Moreover, VSG based photovoltaic power system and their control techniques are presented in various articles [73,96–98]. Various authors proposed different techniques and strategies to tackle the different stability problems in VSG based power systems. For example, [99] proposes a new control strategy based on virtual synchronous generators for an islanded PV-Diesel hybrid generation system to synchronise the output voltages without a phase-locked loop, while correcting the diesel generator rotor initial position and ensuring power-sharing between the two sources, according to the desired power ratio.

Considering the control structure of wind power generation, various researches explore the dynamics of VSG based wind power systems [89,100,101]. For example, in [89] proposes a VSG control for permanent magnet synchronous generator-based wind turbines which enables seamless operation in all operating modes. It also guarantees maximum power point tracking in grid-connected operation, load-following power generation in the islanded operation and low voltage ride-through capability during faults.

6.2. Electric vehicle & wave power systems and VSG

Plug-in electric vehicles have become increasingly commercialised, and vehicle-to-grid technology has received extensive attention. Therefore, few researchers explore the various aspects to stabilise power system by providing necessary power stored in batteries of electric vehicles [102–107]. For example, in [107], a single-phase VSG and possible application for providing vehicle-to-grid services from domestic battery chargers of electric vehicles is presented. The VSG-based electric vehicle charger can contribute to the spinning reserve and the frequency regulation of the power system. In the case of grid outages, the VSG can seamlessly establish an islanded grid and supply local loads from the battery onboard the electric vehicle. Moreover, to avoid the influence on the virtual inertia from power oscillations associated with a single-phase circuit, the proposed control scheme relies on a virtual two-phase system for calculating active and reactive powers. Consequently, the results demonstrate the significant effects of the proposed control in islanded mode and grid connect mode.

Subsequently, for electric power generation from wave energy and connect it to the load or power grid, various scientists do significant research [108–110]. In [110], authors propose a wave power generator with virtual synchronous generator (VSG) control strategy on the inverter side, which can achieve the effect of smoothing the power generation frequency to suppressing power fluctuation. Similarly, in [109] a wave energy converter as the power source is used and power management of VSG is implemented which control the frequency deviation and the terminal voltage.

6.3. AC & DC transmission lines and VSG

The voltage source converter based high voltage alternating current transmission lines is quite common, whereas, high voltage direct current transmission lines are relatively a new trend. According to the current trend, offshore windfarms consists of the high voltage direct current transmission lines, which are entirely connected and controlled by electronic switching devices [111–113]. Due to the higher capacity of offshore windfarms and lack of inertia in it causes critical issues to the power system. Therefore, plenty of researchers applied the VSG control into AC and DC transmission lines [114–121]. In [117], the VSG strategy enables the inverter stations to damp the power oscillation and provide frequency support for the AC system. While, virtual inertia coefficient is obtained to couple the DC voltage and the angular frequency, and inverter station adjust the output power to provide the oscillation damping and frequency support.

In [120,122], researchers proposed an improved control strategy based on the slope control of the DC voltage and AC frequency, in which the virtual inertia is introduced. This strategy ensures that each AC sub-system shares the primary frequency control function and it is easy to apply the secondary frequency control method of traditional AC systems to AC/DC hybrid systems to achieve the steady control of the DC voltage and AC frequency of the whole system. Moreover, in [121], a novel control strategy is proposed to form a multi-terminal low-frequency AC system as an alternative solution to both multi-terminal high voltage direct current and high voltage AC systems in order to provide long distance transmission to interconnect remote power systems. The proposed strategy enables proper power-sharing, synchronisation of each terminal, and frequency stabilisation.

6.4. Energy storage system and VSG

Specifically, the VSGs of the photovoltaic power system can exhibit similar frequency dynamic response as SG. However, unlike SG, power converters are incapable of absorbing/delivering any kinetic energy, which necessitates extra energy storage systems. Therefore, the implementation and coordination control of the energy storage system in VSGs of the photovoltaic power system is necessary, and it has been

investigated by quite a few researchers in [20,123–127].

In [128,129], the authors consider a power system consists of a power electronic converter supplied by a battery bank, which is used to form the AC grid, a wind turbine with its respective power electronic converter and load. The main objective of the proposed strategy is to control the state of charge of the battery bank, limiting the voltage on its terminals by controlling the power generated by the energy sources. Moreover, the electrical frequency of the microgrid is used to inform to the power sources and their respective converters the amount of power they need to generate in order to maintain the battery-bank charging voltage below or equal its maximum allowable limit.

In [130,131] a hybrid energy storage system consisting of a battery and an ultracapacitor is proposed to achieve the power management of VSGs. Through a suitable control, the ultracapacitor automatically tackles the fast-varying power introduced by inertia emulation while the battery implements the remaining parts of a VSG and only compensates for relatively long-term power fluctuations with slow dynamics. The results of this research show that the proposed hybrid energy storage system allows a significant reduction of the battery power fluctuations along with its changing rate.

7. VSG problems and future research directions

The expansion and integration of DGs are increasing rapidly. Therefore, VSG control and related strategies made significant progress in the last decade. Accordingly, the author is mainly engaged in the research of transient stability analysis for distributed power generating systems. Considering VSG control, the author focused on the transient stability improvement according to the centre of inertia and equal area criterion and proposed modification in VSG control. However, in-depth knowledge and significant literature review show that there are plenty of issues related to VSG integration are need to investigate deeply.

7.1. Centralised control for VSG

Currently, the secondary control of VSG based power system can be divided into three types, as centralised control, decentralised control and distributed control. For the centralised control, the microgrid centre controller collects the operational information of the power microgrid and supply it to the optimisation unit of the VSG primary control layer. The centralised method has high adjustment precision due to a central node. However, the reliability and scalability of centralised control are quite weak. In [132], presented a method of microgrid frequency and voltage secondary adjustment using centralised control. The frequency and voltage regulation of the secondary controller is assigned to the VSG's frequency and voltage controller according to the capacity ratio. However, since there is a certain amount of error in the measurement of the power meter, or the frequency adjustment of the frequency adjustment unit, the frequency deviation and the voltage deviation after the adjustment is completed.

The current power system is developing and integrating a large number of VSG based DGs therefore; it is mandatory to develop a centralised control method which improve the different control methods of VSG, such as grid connection, frequency and voltage control, active or reactive power control, and parallel circulation control. To achieve the more stable centralised control for VSG, it is necessary to develop centralised control on the distributed control characteristics of VSG [133–135].

7.2. Reliable and effective modelling of VSG

It is utmost essential to develop reliable and effect modelling techniques for VSG by which a large number of VSGs can effectively integrate into the power system without changing power system crucial parameters. Therefore, a detailed, reliable and accurate dynamic model of VSG and power system is required to develop and analyse deeply. The parameter estimation for VSG and power system should be precise and accurate by some specialised and developed algorithms [31].

7.3. Standardised process to analyse and evaluate VSG performance

The present knowledge of VSG is mostly based on comparative analysis of VSG and SG, whereas the independent understanding of VSG is quite low. Therefore, it is still a problem of how to analyse and evaluate the performance of VSG. Generally, the performance evaluation system is considered as the reflection of all the variables in the system.

Consequently, the important characteristics of VSG are virtual inertia, damping and safe and stable area. Thus, it is crucial to consider these variable while developing the standardised VSG performance analysing and evaluating the system.

7.4. Stability of VSG based power system

As mentioned in previous sections that the stability problem of the VSG based power system have the utmost importance. The increasing number of DGs into power system enhances the stability issue. Although VSG supplies the inertia to the power system and improves the power system coupling, however, the overall stability of power system drops. In the currently available literature, the primary focus of VSG stability problem solving is on individual parts of the system. Whereas the stability of the overall system is not researched in depth. Therefore, for future research, it is necessary to develop the control which possesses the features to improve the stability of individual parts as well as the overall system.

7.5. Standardised interface for VSG

In order to develop the system model, it is crucial to develop a standard model interface which can be used for simulation, control, operation and analysis. The present VSG possess the features of a direct plugin system. However, it has a significant challenge to integrate a large number of VSGs, which make it quite difficult to integrate multiple VSGs into power system [136,137]. Therefore, it is mandatory to develop a standardised interface by which multiple VSGs can integrate into the power system without modifying the fundamental and crucial parameters of the power system. Moreover, different electrical standards of different regions and country should be compatible with the newly developed standard of VSG.

7.6. Standardised interface for parallel connected SG and VSG

The present power system consists of a large number of conventional SG, whereas the VSG is integrating into the same power system with a rapid speed. Consequently, there is some difference in SG and VSG power-sharing to the system during a fault. Generally, VSG quickly injects the power into the system during fault to stabilise the power system frequency, while, the SG release power into the system slowly, which causes the power imbalance in the system. In order to avoid this power imbalance which is caused by SG and VSG power-sharing techniques, it is compulsory to develop a standardised interface for SG and VSG. The interface standards for SG and VSG should be all-inclusive, clear and unambiguous to avoid any misconception.

7.7. Energy storage system for VSG

Generally, the batteries and capacitors are used as an energy storage system with photovoltaic based VSGs. However, batteries and conventional capacitor are allowed to release power; therefore, [128] proposed an ultracapacitor based energy storage system which significantly minimises the release time of stored energy. Moreover, the combination of battery and ultracapacitor is proposed in [130,131],

which suppress the effect of high frequency because ultracapacitor release stored energy rapidly while the low-frequency effect is managed by batteries. Although this solution is quite better for energy storage but not economical due to the high cost. Therefore, it is needed to develop a new and economical type of energy storage system which own the properties of conventional batteries and ultracapacitor and small size.

8. Conclusion

The VSG solves the problem of low inertia and under damping up to some extends. The development of VSG provides a convenient and economical solution to utilise renewable power generation sources and widen their application prospects. Moreover, the growing integration of VSG into power system depicts that a large part of the future power system consists of a large number of VSGs along with conventional SGs. Therefore, the interaction between VSGs and conventional SGs must be studied further. The effective interaction of VSG and SG is an important measure to realise the optimal allocation of renewable energy resources. Furthermore, VSG effectively integrates the flexibility of power electronic equipment and the operating mechanism of SG. In this paper, an overview of different topologies to virtual inertia is presented along with a detailed description of the VSG structure. Moreover, VSG control defining active and reactive power control, and voltage and frequency control is explained in detail. The VSG stability analysis methods, such as small signal stability and transient stability, are presented in comprehensive manners. Later on, the VSG application fields are described. Finally, the VSG problems and future research directions such as centralised control, stability improvement, reliable modelling, interface for VSG etc are discussed.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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